MKN: Metakernel Networks for Few Shot Remote Sensing Scene Classification

Zhenqi Cui, Wang Yang, Li Chen, and Haifeng Li, *Member, IEEE*

***Abstract*— Few-shot remote sensing scene classification tries to make a model quickly adapt to new scenes with only a few samples that do not appear in the closed training set. Since limited samples can hardly describe the distribution of data, it is a challenge for a model to learn good generalized features. Since limited samples are rarely representative, it is another challenge for a model to learn classification boundaries that depend on sample bias. Therefore, we propose a method called metakernel networks (MKNs) to solve the challenges via integrating a parametric linear classifier (PLC) into the metalearning framework to address the former problem assembling a metakernel strategy (MKS) and a stretching loss (Stloss) to address the later problem. The PLC learns prior knowledge from different tasks sampled from the same task family to learn rich features. The MKS is designed to remap low-dimensional indistinguishable features to a high-dimensional space to solve the low-dimensional feature entanglement caused by large intraclass differences between remote sensing images. The Stloss reduces the dependence of the hyperplane formed by a few points in each class on sample selection by reducing the intraclass and interclass variance ratios, thus solving the boundary fragility problem caused by interclass similarity. Experiments on three public datasets, UC_Merced, NWPU-RESISC45, and AID, show the state-of-the-art performance by the accuracy improvement of 5.04%, 4.81%, and 4.69% respectively. Our findings provide a new perspective by suggesting that the previously neglected issue of classification boundaries may be a key factor for few-shot remote sensing scene classification.**

***Index Terms*— Few-shot learning (FSL), remote sensing scene classification.**

I. Introduction

THE aim of scene classification, a fundamental and important problem in the field of remote sensing image interpretation, is to assign a specific semantic label to each untagged remote sensing image based on its semantic content [1]–[5]. The task has a wide range of applications, including disaster detection, housing planning [4], [6], environmental monitoring, and land resource management [7]–[10].

In recent years, deep learning-based methods have achieved many excellent results in remote sensing scene classification due to their excellent feature extraction capabilities and the development of high-performance computing units [11]–[17]. The essence of this class of methods is the end-to-end extraction of features using deep neural networks, such as autoencoders (AEs) [18], deep belief networks (DBNs) [19], and convolutional neural networks (CNNs) [20]. On the one hand, these methods are based on closed world assumptions; however, in reality, the world is open, and we have to face new remote sensing scenarios that do not exist in closed datasets [21]–[23]. On the other hand, they are also data hungry; therefore, these standard supervised deep learning paradigms often cannot provide a satisfactory solution for learning from few data due to overfitting [24]–[26]. Therefore, to reduce the reliance on data annotation, more researchers have strived to develop powerful methods for learning new concepts from very few samples, which is known as few-shot learning (FSL) [27]–[33].

To address the FSL problem, the state-of-the-art methods include metalearning, whose core concept is to learn a model that, given a task (a set of labeled data), produces a classifier that can be generalized across all tasks [34], [35]. By introducing metalearning strategies and metric learning frameworks, Koch *et al.* [36] and Vinyals *et al.* [37] contributed to early small sample classification methods. Snell *et al.* [30] and Sung *et al.* [31] built on this foundation with further improvements. This class of methods usually consists of an embedding module and a metric module. For a given unlabeled remote sensing image and several labeled images, the embedding module first generates low-dimensional embeddings of all the input images, and then, the metric module gives recognition results by measuring the distance between the unlabeled image features and the sample image features. The essence of this class of methods is the expectation that the constructed embedding space can better fit some prespecified distance metric function, such as cosine similarity [37] or Euclidean distance [30]. In practice, however, since only the embedding module is learnable, the fixed metric function limits the feature extractor from producing a discriminative representation. Sung *et al.* [31] attempted to address the above problem by introducing an implicitly learnable metric module, but it has been shown that the comparison ability is still limited due to its inherent local connectivity [38]. Therefore, the first central concept of our approach is to introduce a parameterized linear classifier that allows more flexibility into the metametric learning framework instead of a nonlearnable fixed metric function since the former can produce a different set of parameters for different tasks and thus learn more task-specific knowledge, thus better compensating

Manuscript received August 14, 2021; revised October 27, 2021 and January 11, 2022; accepted February 8, 2022. Date of publication February 22, 2022; date of current version March 31, 2022. This work was supported by the National Natural Science Foundation of China under Grant 41571397, Grant 41871364, Grant 41671357, and Grant 41871302. *(Corresponding author: Haifeng Li.)*

The authors are with the School of Geosciences and Info-Physics, Central South University, Changsha 410083, China (e-mail: lihaifeng@csu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3153679

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

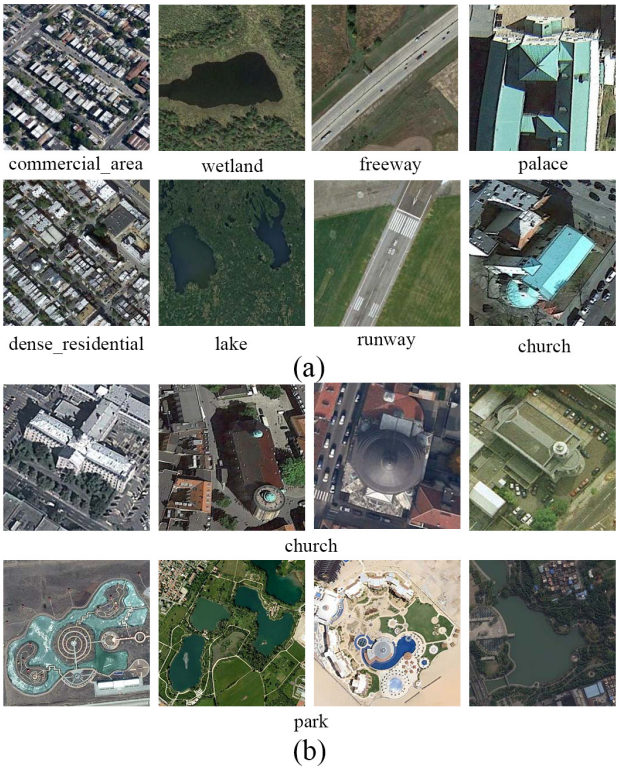


Fig. 1. (a) Interclass similarity of remotely sensed images. (b) Intraclass variability of remotely sensed images. The effects of both phenomena are magnified when the sample size is too small. The above remote sensing images are taken from the public datasets AID and NWPU-RESISC45.

for the lack of features learned due to the small sample size.

Furthermore, scene images from different semantic categories of remote sensing imagery often have a high degree of similarity. As shown in Fig. 1(a), "commercial_area," "dense_residential," "wet land and lake," "freeway," and "runway" are almost indistinguishable, and "palace" and "church" are also very similar. The scarcity of labeled samples amplifies the effect of these similar individuals on classification boundaries, making the high similarity between these categories lead to entanglement of the low-dimensional features generated by the embedding module, which prevents the metric module from performing an effective linear division. Therefore, the second core concept of our approach is that we propose a metakernel strategy (MKS) to quadratically map the low-dimensional features to a higher dimensional space, disentangle the entangled features from a higher dimension, improve the clarity of the classification boundaries of different categories, and thus improve the classification results.

In addition, due to their complexity and diversity, ground objects are also affected by imaging conditions, such as background, lighting, and scale during the capture process. As a result, as shown in Fig. 1(b), there are large visual differences between objects in scene images of the same semantic class, resulting in large intraclass variation in remotely sensed images. This large intraclass variation causes the intraclass feature dispersion phenomenon. When the number of samples is sufficiently rich, the large number of samples compensates for the dependence of class boundaries on the sample. However, in the case of small samples, when the features in a class become dispersed, the intraclass-to-interclass variance ratio rises, and the dependence of the class boundary on the selection of one sample becomes stronger. The classification boundary formed by sampling one data point at this time tends to misclassify large regions, thus affecting the overall classification accuracy. Therefore, the third core concept of our approach is to propose stretching loss (Stloss), which minimizes the dependence of class boundaries on sample selection and improves the robustness of classification boundaries by constraining the intraclass-to-interclass variance ratio and forcing interclass features to aggregate.

In summary, we propose a network framework called metakernel networks (MKNs) to improve the effectiveness of few-shot remote sensing scene classification. Our approach remains within the framework of metametric learning and follows the training strategy of episodes. First, we discard the nonparametric nearest neighbor classifier and incorporate the parametric linear classifier (PLC) into the metric space to remove the limitation of the fixed metric function on the learnability of the model. Parameterized linear classifiers allow more flexibility because they produce a different set of parameters for different tasks and can learn more task-specific knowledge, which can better compensate for the lack of features. Second, we propose the MKS, which maps the features in the metric space to the high-dimensional space twice, thus effectively solving the low-dimensional feature entanglement problem caused by the high similarity between remote sensing image classes. Third, we propose a new loss function named Stloss. This loss function reduces the dependence of category boundaries on sample selection by constraining the intraclass and interclass variance ratios, thus effectively addressing the phenomenon of category misclassification caused by the great interclass variability of remotely sensed images. We evaluate the proposed approach on three public benchmark datasets, and the experimental results show that our proposed MKN method outperforms existing methods and achieves state-of-the-art results.

In summary, the main contributions of our proposed model can be summarized as follows.

1) By effectively incorporating PLCs into a metalearning framework, our model makes efficient use of task-specific information and lifts the limitation of fixed metrics on the model.
2) We propose an MKS to quadratically map the features in the embedding space to the high-dimensional space, which solves the low-dimensional feature entanglement problem caused by the interclass similarity of remote sensing images, improves the classification boundary clarity, and enhances the classification effect.
3) We propose Stloss, which minimizes the dependence of category boundaries on sample selection by reducing the intraclass to interclass variance ratio, effectively increasing the robustness of the classification boundaries, and improving the model classification effect.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

4) Experimental results on three open and challenging remote sensing datasets demonstrate that our proposed MKN method achieves state-of-the-art results.

The rest of this article is organized as follows. In Section II, we introduce works related to few-shot remote sensing scene classification. Section III describes the proposed method. We describe the experiments in Section IV, and finally, conclusions are presented in Section V.

II. Related Work

The core concept of FSL is rapid generalization to a new task through a few samples with supervised information, and this approach can be broadly divided into three groups: generation-based methods, gradient-based methods, and metric-based methods.

Generation-based methods extend a small dataset to a large dataset by generating fake samples. Hariharan and Girshick [39] proposed representation regularization techniques and techniques to mimic additional training examples for data-starved classes. Yang *et al.* [40] calibrated the distribution of few-sample classes by transferring statistics from the classes with sufficient examples, and similarly, Zhang *et al.* [41] proposed a spectral–spatial fractal residual CNN with data balancing enhancement to address the problem of limited labeled samples in hyperspectral classification. In addition, Bao *et al.* [42] proposed bowtie networks that jointly learn 3-D geometric and semantic representations with a feedback loop.

Gradient-based methods focus on quickly adapting model parameters to new tasks through gradient descent. Finn *et al.* [25] and Zhai *et al.* [44] iteratively selected a batch of previous tasks, trained learners for each task to calculate gradients and losses, and updated in a direction in which the weights are more easily updated by backpropagation. Tseng *et al.* [44] demonstrated the usefulness of the MAML method [24] for creating maps of crop types given a small number of labels. Rußwurm *et al.* [45] proved the superior performance of the MAML algorithm on few-shot problems in land cover classification. Nichol *et al.* [46] performed K iterations of stochastic gradient descent for a given task and then gradually moved the initialization weights toward the weights obtained after K iterations. Ma *et al.* [47] applied parameter shifting to fix the parameters in the DNN to solve the problem of training a large number of parameters. Li *et al.* [26], Zhou *et al.* [49], and Antoniou *et al.* [50] also learned the step size, Wang *et al.* [51], Duan *et al.* [52], and Santoro *et al.* [53] trained the recurrent network to predict the step size from the gradient, and Li *et al.* [53] deployed self-supervised embedding optimization for adaptive generalization in new tasks.

Metric-based methods, to which our method belongs, aim to obtain a feature space that can be adapted to new tasks by means of metric constraints. This feature space coincides with the weights of the network. A new task can be learned by comparing a new input with a label in the feature space. The higher the similarity between a new input and this label, the more likely it is that the new input will belong to the class represented by that label. Koch *et al.* [36] and Vinyals *et al.* [37] built the basic learning paradigm and laid the foundation for such methods. Snell *et al.* [30] built on this by pioneering the concept of prototype and used a feedforward neural network to embed the task examples and perform nearest neighbor classification with the prototype. Furthermore, Li *et al.* [23] used a learnable distance metric and proposed a balance loss with the aim of maximizing the fitting capability while maximizing generalization capability across different remote sensing scenarios. Li *et al.* [54] introduced channel attention and spatial attention modules to obtain more discriminative remote sensing image features, and Zhu *et al.* [55] captured the discriminative feature information of remote sensing scenes from both global and local perspectives by combining multiple attention mechanisms and the attention-reference mechanism. In contrast to these works, our proposed MKN shows that the previously neglected classification boundary problem may be a key factor in a few-shot remote sensing scenes classification. In addition, our proposed approach suggests a targeted solution to a unique and important intraclass and interclass ambiguity problem that has a strong impact on a few-shot remote sensing scenes classification.

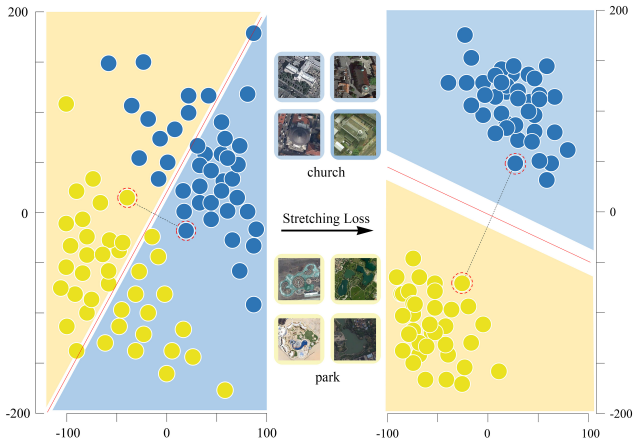


Fig. 2. When the sample size is too small, it amplifies the effects of interclass differences and intraclass similarities in remotely sensed imagery. The formation of classification boundaries is strongly influenced by sample selection.

III. Proposed Method

In this section, we give the details of the proposed MKN for a few-shot remote sensing classification. Our approach follows the training strategy of episodes in the framework of metametric learning. For each task, the embedding module acquires remote sensing image features, and the metric module performs linear classification of the different features. On the one hand, for the low-dimensional feature entanglement problem, we propose an MKS to quadratically map them to a higher dimensional space for untangling; on the other hand, to reduce the dependence of category boundaries on sample selection, we constrain the intraclass and interclass variance ratios by Stloss to force intraclass feature aggregation. An overview of our proposed method is given in Fig. 3.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

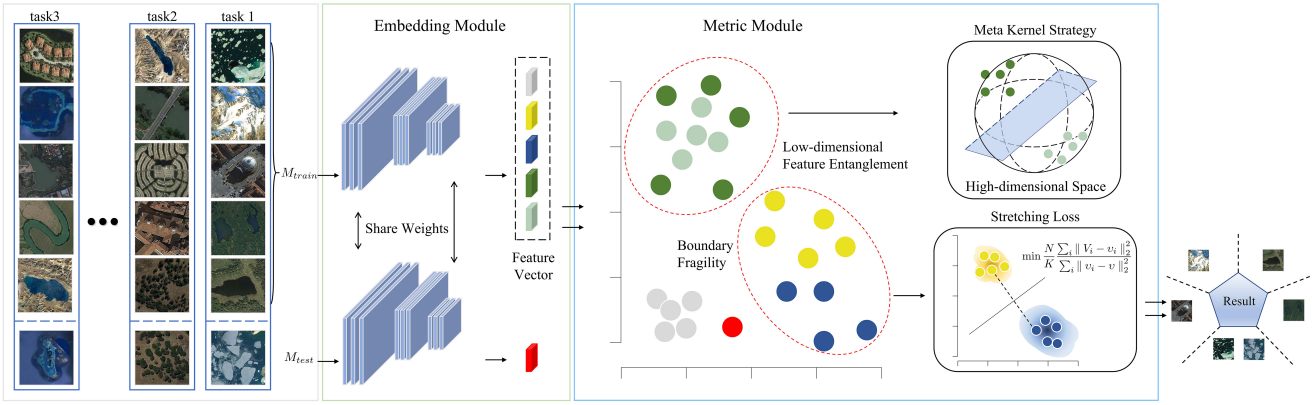


Fig. 3. Overall architecture of our proposed model for a five-way one-shot setting with one query image. For each task, MKN first maps the support images (M_{train}) into embedded features (colored cubes) using the embedding module. Then, the metric module classifies the embedded features by a PLC, which is processed using metakernel and Stloss (see Section III-C for details) for low-dimensional entangled features and boundary fragility problems, respectively. The query set usually has the same number of classes as the support set and is classified using the weights fixed for this task.

A. Problem Definition

Given nonoverlapping remote sensing datasets C_{seen} and C_{unseen}, our goal is to quickly classify N different remote sensing scenes drawn from C_{unseen}, each of which contains only K labeled samples, which is the so-called N-way K-shot setting. Since we are proposing to solve few-shot remote sensing scene classification problems, the setting of K is usually small, typically $K \in \{1, 5\}$. The training set D_{train} and the validation set D_{val} are constructed from C_{seen}, and the test set D_{test} is constructed from C_{unseen}. The validation set D_{val}, which is designed to select the learner's hyperparameters and to choose the best model, does not intersect with D_{train}, that is, $D_{\text{train}} \cap D_{\text{val}} = \emptyset$. D_{train}, D_{val}, and D_{test} all contain multiple tasks/episodes. Each task/episode consists of a support set M_{train} and a query set M_{test}, namely, $D_{\text{train}} = \{M_{\text{train}}, M_{\text{test}}\}_{i=1}^{E_{\text{tr}}}$, $D_{\text{val}} = \{M_{\text{train}}, M_{\text{test}}\}_{i=1}^{E_{\text{val}}}$, and $D_{\text{test}} = \{M_{\text{train}}, M_{\text{test}}\}_{i=1}^{E_{\text{ts}}}$, where $E_{\text{tr}}, E_{\text{val}}$, and E_{ts} represent the number of episodes during training, validation, and testing, respectively. Both M_{train} and M_{test} are image sets. In the training phase, for each episode, we construct the support set $M_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{NK}$ by sampling N different categories from the D_{train}, with K labeled samples per category, and construct the query set $M_{\text{test}} = \{(x_i, y_i)\}_{i=1}^{NK}$ data from the remaining samples from the same N categories, so M_{train} and M_{test} do not have the same remote sensing image, i.e., $M_{\text{train}} \cap M_{\text{test}} = \emptyset$.

B. Embedding Module

Our ultimate goal is to obtain a predictive model $\hat{y} = f_\varphi(x)$ parameterized by φ by solving

$$\varphi^\star = \arg\min_{\varphi} \; L(D_{\text{train}}; \varphi, \omega) \tag{1}$$

where L is a loss function that measures the error between true labels and those predicted by $f_\varphi(\cdot)$. L consists of two parts, which we describe in detail in Section III-D. The task-specific classification model parameterized by ω is described in Section III-C. With the embedding module, we map the data domain to the feature space so that visual information is related to each other, that is, for each episode, a given dataset M_{train} is embedded into a vector

$$V_{M_{\text{train}}} = f_\varphi(M_{\text{train}}; \varphi). \tag{2}$$

C. Metric Module

The metric module needs to give an optimal classification scheme for the feature vector obtained from the embedding module. Traditional methods mostly use nearest neighbor classifiers [30], [37], [56]; however, this nonparametric approach ignores task-specific information, and the fixed metric function limits the feature extractor to produce a discriminative representation. We, therefore, use parameterized linear classifiers to allow greater flexibility in the metric space, as they can generate a different set of parameters for different tasks and learn more task-specific knowledge, i.e.,

$$\mathop{\mathbb{E}}_{\substack{\mathcal{T}\sim p(\mathcal{T}) \\ (M_{\text{train}}, M_{\text{test}})\in\mathcal{T}}} \sum_{y\in M_{\text{test}}} [(\theta^T \cdot V) - y^2]. \tag{3}$$

$\mathcal{T}$ denotes a task sampled during training, and $p(\mathcal{T})$ is the distribution of tasks. θ defines the linear regression weights used to predict the examples in the meta test set, providing a good solution for the new tasks sampled from $p(\mathcal{T})$. For each task, there is

$$\begin{aligned} \theta = \{\omega_n\}_{n=1}^{N} = \arg\min_{\{\omega_n\}} \min_{\{\rho_i\}} \; & \frac{1}{2} \sum_n \| \omega_n \|_2^2 + \eta \sum_{nk} \rho_{nk} \\ \text{s.t. } & \omega_{y_i} \cdot f_\varphi(x_i) - \omega_n \cdot f_\varphi(x_i) \geq 1 - \delta_{y_i,n} - \rho_n \quad \forall i, n \end{aligned} \tag{4}$$

where

$$\delta_{i,j} = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j. \end{cases} \tag{5}$$

Our embedding module is differentiable, and for the whole model to be trainable end-to-end, this requires our metric module to also be differentiable, i.e., we need to calculate the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

gradient of the linear classification model and then backpropagate it to optimize the model. Consider the following convex optimization problem:

$$\begin{aligned} \min\ & f_0(\theta, x) \\ & f(\theta, x) \le 0 \\ & h(\theta, x) = 0. \end{aligned} \tag{6}$$

Here, vector θ is the optimization variable of the problem and vector x is the input data, which in our case is the embedding vector V. Function $f_0(\cdot)$ is the objective function, and functions $f(\cdot)$ and $h(\cdot)$ are the equation constraint and inequality constraint, respectively. Assuming that function f_i is convex at a fixed θ, we can optimize the objective by solving for the saddle point (θ, $\tilde{\lambda}$, and $\tilde{\nu}$) of the following Lagrangian:

$$F(\theta, \lambda, \nu, x) = f_0(\theta, x) + \lambda^T f(\theta, x) + \nu^T h(\theta, x). \tag{7}$$

According to [57], a set of functions is functionally related if the Jacobi determinant is constantly equal to zero in a connected region, where at least one of the functions is a continuously differentiable function of the rest, i.e., the Jacobi matrix can embody the optimal linear approximation of a differential equation to the given point. Therefore, we let $z = (\theta, \lambda, \nu)$ and define function

$$g(z, x) = \begin{bmatrix} \nabla_\theta F(\theta, \lambda, \nu, x) \\ \mathbf{diag}(\lambda) f(\theta, x) \\ h(\theta, x) \end{bmatrix} \tag{8}$$

where $\mathbf{diag}(\cdot)$ denotes the diagonalisation operation. We can then define the Jacobian matrix. The optimal solution is obtained when and only when $g(\tilde{z}, x) = 0$ (see the Appendix for detailed proof process). When we compute the optimal solution $\tilde{\theta}$, we can obtain the gradient-optimal linear approximation, which allows backpropagation, thus ensuring that the model as a whole can be trained end-to-end.

For low-dimensional entangled features, we introduce the MKS, which uses a nonlinear mapping $\Phi(x)$ to quadratically map the features in the metric space to the high-dimensional space $D = \mathbb{R}^C$ (C can be infinite) and increases the separability between different classes by expanding the spatial resolution around the surface of the separation boundary through a conformal mapping, with the kernel function defined as follows.

$$\widetilde{K}(x, x') = c(x)c(x')K(x, x') \tag{9}$$

where

$$K(x, x') = e^{-\|x-x'\|^2/2\sigma^2}. \tag{10}$$

Also,

$$c(x) = \sum_{i \in NK} r_i e^{-\|x-x_i\|^2/2\sigma^2} \tag{11}$$

where although σ is the kernel function hyperparameter, it can be learned together with θ by different tasks, all without human setting, which not only allows the model to run in a data-driven manner but also greatly improves the effectiveness and efficiency of the model. At this point, (3) can be reexpressed as

$$\mathop{\mathbb{E}}_{(M_{\text{train}}, M_{\text{test}}) \in \mathcal{T}} \sum_{y \in M_{\text{test}}} [(\theta^T \cdot \Phi(V) - y^2]. \tag{12}$$

D. Loss Function

Our loss function consists of two parts, the first of which measures the classification effect in the embedding module. We evaluate the negative log likelihood of test data sampled from the same distribution

$$L_{\text{cls}} = \sum_{x, y \in M_{\text{test}}} -\left[\omega_y \cdot \Phi(f_\varphi(x)) - \log \sum_k e^{\omega_k \cdot \Phi(f_\varphi(x))} \right]. \tag{13}$$

The second part we named Stloss. The great similarity between classes in remotely sensed imagery causes intraclass feature dispersion in the metric space, and when the data volume is small, the classification boundary is easily affected by the sampling points, i.e., the classification boundary is fragile. Therefore, to reduce the dependence of class boundaries on sample selection and improve the robustness of classification boundaries, we propose Stloss, which forces intraclass feature aggregation by constraining the intraclass and interclass variance ratios

$$L_{\text{str}} = \frac{N}{K} \frac{\sum_i \| V_i - v_i \|_2^2}{\sum_i \| v_i - v \|_2^2}. \tag{14}$$

Therefore, our loss function is

$$L = L_{\text{cls}} + \alpha L_{\text{str}} \tag{15}$$

where α is the hyperparameter that controls the weights, and our extensive experiments on the validation set show that the model achieves the best performance when $\alpha = 0.2$.

IV. Experiment and Analysis

In this section, we evaluate the proposed MKN method on three public datasets designed for remote sensing scene classification. We first introduce the dataset used in the experiment (Section IV-A). Then, we describe the network architecture and hyperparameter design in detail (Section IV-B). Finally, we analyze our model and present our experimental results on different datasets (see Sections IV-C–IV-E).

A. Dataset Description

In the experiments, we evaluate the proposed MKN method on three publicly available datasets designed for remote sensing image scene classification.

The NWPU-RESISC45 dataset [58] contains a total of 31 500 images divided into 45 scene categories: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED DIVISIONS ON THREE DATASETS

Datasets	Training Set	Validation Set	Test Set
NWPU-RESISC45	Airplane; BaseballDiamond; Beach; Bridge; Chaparral; Church; Cloud; Desert; Freeway; GolfCourse; Harbor; Island; Lake; Meadow; MobileHomePark; Mountain; Palace; Railway; RectangularFarmland; Roundabout; SeaIce; Ship; SparseResidential; Stadium; Wetland;	ThermalPowerStation; Overpass; RailwayStation; TennisCourt; CommericalArea; Snowberg; StorgeTank; Runway; Terrace; IndustrialArea;	CircularFarmland; BasketballCourt; Forest; DenceResidential; River; GroundTrackField; Intersection; MediumResidential; ParkingLot; Airport;
AID	Airport; Bareland; Baseballfield; Beach; Bridge; Commercial; DenseResidential; Desert; Farmland; Meadow; Pond; Port; Railwaystation; Resort; Stadium;	MediumResidential; Mountain; Park; Parking; Playground;	River; School; SparseResidential; Square; Forest; Industrial; Center; Church; StorageTanks; Viaduct;
UCMerced_LandUse	Agricultural; BaseballDiamond; Buildings; Chaparral; DenseResidential; Freeway; Harbor; MediumResidential; Overpass; Parkinglot;	Airplane; Forest; Intersection; Runway; StorageTanks;	SparseResidential;GolfCourse; Mobilehomepark; Tenniscourt; River; Beach;

overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. Each category consists of 700 images in the red–green–blue (RGB) color space and is 256 × 256 pixels in size. For most scene categories, the spatial resolution varies from approximately 0.2 to 30 m per pixel. To the best of our knowledge, this dataset is the largest in terms of the number of scene categories and the total number of images, with rich image variation, large intracategory diversity, and high intercategory similarity, making the dataset more challenging.

The UCMerced_LandUse dataset [59] contains 21 kinds of land use categories: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. For each scene category, there are 100 aerial images in the RGB color space, with a size of 256 × 256 pixels and a spatial resolution of 0.3 m. Since its emergence, this dataset has been widely used for remote sensing scene classification.

AID [60] is a large-scale dataset used for aerial scene classification including agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. It contains a total of 10 000 images in 30 scene categories with a size of 600 × 600 pixels. The number of images varies from 220 to 420 under different aerial scene categories. The spatial resolution varies from approximately 0.5 to 8 m.

The training set, validation set, and test set are divided, as shown in Table I.

B. Network Architecture and Hyperparameter Design

For fairness, our initial embedding model has four modules with 3 × 3 convolutions and 64 filters, with padding = 1 to retain as much feature information as possible, followed by batch normalization [61], ReLU nonlinearity [62], and 2×2 max pooling. How the embedded model affects the experimental results is shown in Section IV-C. All remote sensing images are resized to 256 × 256 and we did not perform image enhancement processing. During the training phase, the weight decays every 20 epochs, and the corresponding parameter is set to 0.0005. The learning rate is set to 0.001. For the loss function, α is set to 0.2.

TABLE II
CLASSIFICATION ACCURACY (%) OF THE DIFFERENT METHODS FOR THE NWPU-RESISC45 DATASET (AVERAGE OF 20 RUNS ± STANDARD DEVIATION; THE BOLD VALUES REPRESENT THE BEST ACCURACY AMONG THESE METHODS IN EACH CASE

Method	5-way	
	1-shot	5-shot
Matching net [37]	37.28 ± 0.99	52.31 ± 0.73
MAML [24]	42.29 ± 0.76	61.85 ± 0.81
Prototypical network [30]	43.99 ± 0.78	69.35 ± 0.66
Meta-SGD [25]	43.98 ± 0.80	63.14 ± 0.55
LLSR [43]	42.20	62.01
Relation network [31]	44.55 ± 0.79	68.21 ± 1.39
RS_MetaNet [23]	52.78 ± 0.09	71.49 ± 0.81
DLA-MatchNet [54]	60.80 ± 0.70	77.63 ± 0.46
MKN(ours)	**65.84 ± 0.89**	**82.67 ± 0.55**

C. Classification Accuracy

We evaluate experimental results using classification accuracy (acc), which is defined as

$$\text{Acc} = \frac{1}{E_{\text{ts}}} \sum_{i=1}^{E_{\text{ts}}} \frac{r^{(i)}}{NK} \tag{16}$$

where $r^{(i)}$ denotes the number of correctly classified samples in task i. Also, we compare the proposed MKN with the representative/state-of-the-art methods on the three benchmark datasets in Tables II and III. It is easy to see that whether given only one sample ($K = 1$) or five samples ($K = 5$), our method shows the best performance and achieves 57.29% and 75.42% scene classification accuracy on the UC_Merced dataset, 58.45% and 77.92% scene classification accuracy on the AID dataset, and 65.84% and 82.67% scene classification accuracy on the NWPUSISCS dataset. In addition, we report the results measured in a confusion matrix on each of the three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

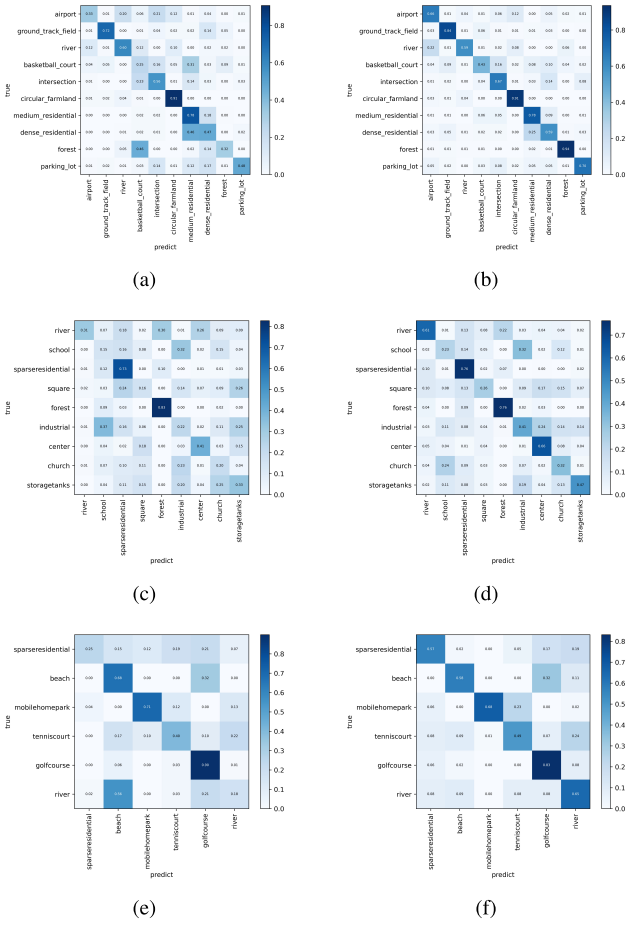


Fig. 4. Confusion matrices of the (Top) NWPU-RESISC45 dataset, (Middle) AID, and (Bottom) UCMerced_LandUse dataset. Results for one-shot are on the left and five-shot ones are shown on the right. (a) One-shot on NWPU. (b) Five-shot on NWPU. (c) One-shot on AID. (d) Five-shot on AID. (e) One-shot on UCM. (f) Five-shot on UCM.

TABLE III

CLASSIFICATION ACCURACY (%) OF THE DIFFERENT METHODS FOR THE AID DATASET (AVERAGE OF 20 RUNS ± STANDARD DEVIATION; THE BOLD VALUES REPRESENT THE BEST ACCURACY AMONG THESE METHODS IN EACH CASE

Method	5-way	
	1-shot	5-shot
Matching net [37]	41.53 ± 0.54	50.04 ± 0.34
MAML [24]	47.93 ± 0.72	61.79 ± 0.75
Prototypical network [30]	49.80 ± 0.56	68.64 ± 0.29
Meta-SGD [25]	49.97 ± 1.23	64.84 ± 0.97
LLSR [43]	48.57	62.1.45
Relation network [31]	53.51 ± 0.68	67.39 ± 1.04
RS_MetaNet [23]	56.32 ± 0.55	74.48 ± 1.11
DLA-MatchNet [54]	53.41 ± 0.99	70.61 ± 0.36
MKN(ours)	**57.29 ± 0.59**	**75.42 ± 0.31**

datasets. In each dataset, we show the one-shot and five-shot confusion matrices, as shown in Fig. 4, where the entry in the ith row and jth column represents the rate of test images from the ith class that are classified as the jth class.

Furthermore, we used the four-layer convolution, vgg16, resnet12, resnet50, and resnet152 as the backbone network of our embedding module and present the results in Fig. 5.

TABLE IV

CLASSIFICATION ACCURACY (%) OF THE DIFFERENT METHODS FOR THE UCMERCED_LANDUSE DATASET (AVERAGE OF 20 RUNS ± STANDARD DEVIATION; THE BOLD VALUES REPRESENT THE BEST ACCURACY AMONG THESE METHODS IN EACH CASE

Method	5-way	
	1-shot	5-shot
Matching net [37]	42.22 ± 1.02	52.32 ± 0.92
MAML [24]	47.53 ± 0.71	63.13 ± 0.92
Prototypical network [30]	50.23 ± 0.61	70.54 ± 0.20
Meta-SGD [25]	48.24 ± 0.42	65.54 ± 0.41
LLSR [43]	47.95	63.32
Relation network [31]	52.43 ± 0.87	68.65 ± 0.95
RS_MetaNet [23]	57.23 ± 0.56	76.08 ± 0.28
DLA-MatchNet [54]	53.76 ± 0.62	63.01 ± 0.51
MKN(ours)	**58.45 ± 0.54**	**77.92 ± 0.32**

TABLE V

CLASSIFICATION ACCURACY (%) OF THE DIFFERENT KERNEL FUNCTIONS ON THE NWPU-RESISC45 DATASET

Kernel Function	5-way		
	1-shot	5-shot	10-shot
Linear Kernel	50.25	66.54	67.02
Polynomial Kernel	53.24	69.19	69.56
Gaussian Kernel	60.29	77.42	78.16
ours	65.84	82.67	84.23

For the four-layer convolution, vgg16 and resnet12, the model performance was positively correlated with the depth of the backbone, indicating that the enhanced feature extraction capability of the embedding module could improve the overall performance of the model. However, when resnet50 and resnet152 are used as the backbone, the more powerful feature extraction capability does not bring further improvement but instead causes a significant drop in model performance due to the meta overfitting phenomenon caused by the overly deep network, which will be discussed in detail in the discussion module.

D. Metric Space Analysis

As mentioned earlier, when the number of samples is too small, it will greatly increase the dependence of the classification boundary on sample selection, which in turn exacerbates the effects caused by intraclass variation and interclass similarity in remote sensing images, generating the problems of low-dimensional feature entanglement and fragile classification boundaries. We propose an MKS and a Stloss to address these problems. To better illustrate this phenomenon and how our model works, we used the T-sne [63] algorithm to visualize the results of MKN and relation network in a reduced dimension, as shown in Fig. 6. It is clear that the MKN algorithm effectively alleviates the intraclass feature entanglement problem and increases the separation between classes.

For the MKS, we further analyzed the effect of different kernel functions on the model, and the results are shown in Tables V–VII. The results show that our strategy shows outstanding results compared to a linear kernel function,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

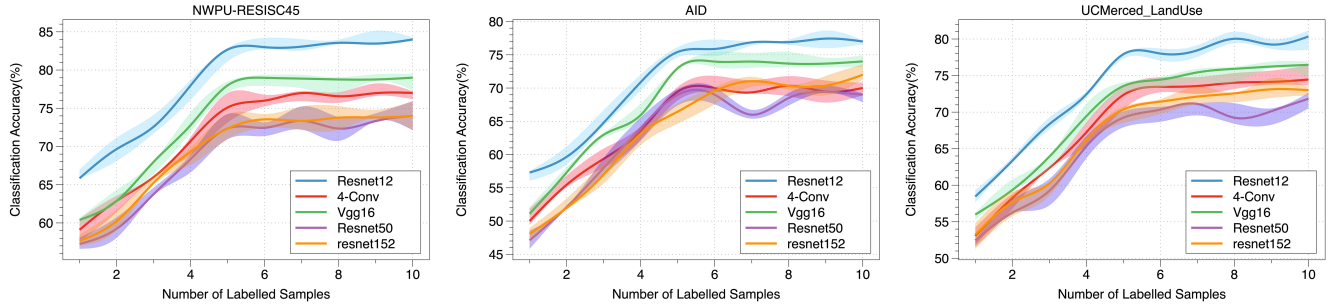


Fig. 5. Classification accuracy of different backbones on three datasets. The horizontal axis represents the number of labeled samples given on D_{test}.

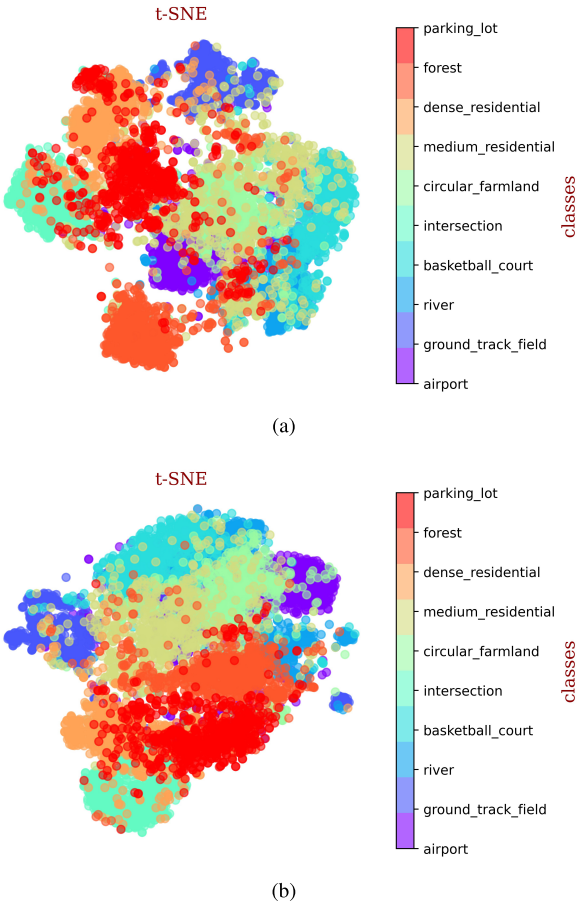


Fig. 6. Visualization results of (a) MKN and (b) RelationNetwork on the NWPU-RESISC45 dataset by T-sne with only one labeled sample.MKN makes the space more distinctive.

TABLE VI

CLASSIFICATION ACCURACY (%) OF THE DIFFERENT KERNEL FUNCTIONS ON THE AID DATASET

Kernel Function	5-way		
	1-shot	5-shot	10-shot
Linear Kernel	40.89	55.76	56.54
Polynomial Kernel	45.56	59.87	61.45
Gaussian Kernel	53.34	69.76	70.01
ours	57.29	75.42	76.97

polynomial kernel function, and Gaussian kernel function. This is mainly because our strategy can increase the separability between different classes by expanding the spatial resolution around the surface of the separation boundary through conformal mapping.

TABLE VII

CLASSIFICATION ACCURACY (%) OF THE DIFFERENT KERNEL FUNCTIONS ON THE UCMERCED_LANDUSE DATASET

Kernel Function	5-way		
	1-shot	5-shot	10-shot
Linear Kernel	44.68	64.76	65.22
Polynomial Kernel	45.55	65.35	66.25
Gaussian Kernel	46.22	67.61	69.16
ours	58.45	77.92	79.43

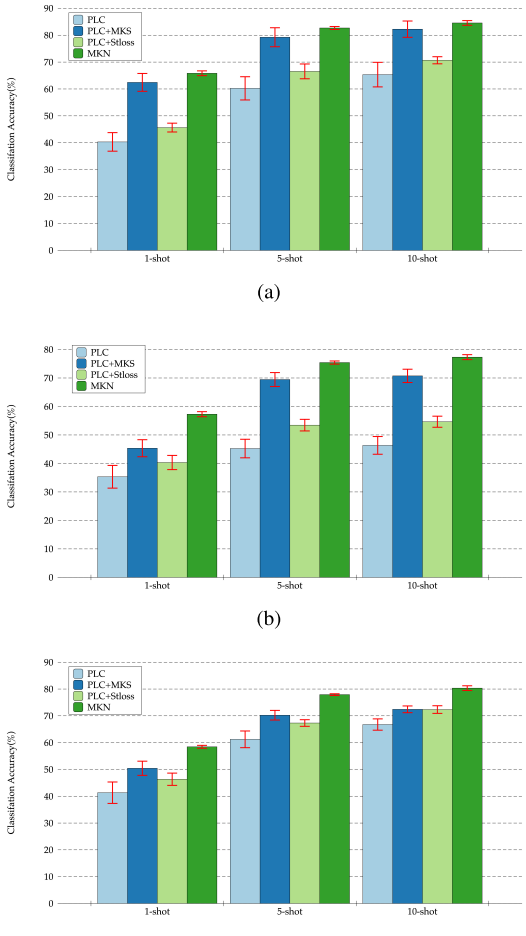


Fig. 7. Ablation study with the MKS and Stloss removed on three datasets. (a) NWPU-SISCS45. (b) AID. (c) UCMerced_LandUse.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

E. Ablation Study

In this section, we investigate the effect of the MKS and Stloss on the generalization accuracy. We show the results for five-way one-shot, five-shot, ten-shot on three datasets, UCMerced_LandUse, AID, and NWPU-RESISC45. The results demonstrate the great usefulness of the MKS. When only one labeled sample is available, the dependence of the model on sample selection is greatly enhanced, which amplifies the intraclass variation and interclass similarity nature of remote sensing images to a considerable extent, which makes the low-dimensional feature entanglement phenomenon more prevalent and its impact more significant. The MKS, on the other hand, is a good solution to the above problems. In addition, the Stloss also has a positive effect on the overall performance, although not as pronounced as the metakernel effect. This may be due to the tradeoff between linear classification loss and Stloss.

V. Conclusion and Future Work

In this article, we propose a framework called the MKNs to solve the few-shot remote sensing scene classification problem. First, to address the problem of underlearning features due to small samples, we propose to fuse linear classifiers into the metametric learning framework so that the model can not only learn prior knowledge from different tasks to compensate for inadequate features but also make the learned knowledge more task-specific. Second, we propose an MKS to solve the low-dimensional feature entanglement problem by quadratically mapping low-dimensional features to high-dimensional features. Finally, we propose stretch loss, which improves the class separation by constraining the intraclass to interclass variance ratio so that the hyperplane formed by selecting one point in each class does not vary significantly with sample selection. We conduct comparative experiments on several publicly available remote sensing image datasets, and the experimental results show the excellent performance of MKN in scene classification tasks. As mentioned in Section III, our model still suffers from overfitting in the metatraining phase, which wastes the feature extraction capability of the deeper network; therefore, our future work will focus on how to exploit the deeper network more efficiently without meta overfitting. In conclusion, our MKN method provides an effective methodological reference for the small sample scene classification problem.

Appendix

Consider the following convex optimization problem:

$$\begin{aligned} \min\ & f_0(\theta, x) \\ & f(\theta, x) \preceq 0 \\ & h(\theta, x) = 0. \end{aligned} \tag{17}$$

Define the Lagrangian

$$F(\theta, \lambda, \nu, x) = f_0(\theta, x) + \lambda^T f(\theta, x) + \nu^T h(\theta, x). \tag{18}$$

Then, necessary and sufficient optimality conditions for (17) are given as follows. The optimal solution is obtained if and only if there are $\tilde{\lambda}$ and $\tilde{\nu}$, which together with $\tilde{\theta}$ satisfies the KKT conditions

$$\begin{aligned} f(\tilde{\theta}, x) &\preceq 0 \\ h(\tilde{\theta}, x) &= 0 \\ \tilde{\lambda}_i &\geq 0 \\ \tilde{\lambda}_i f_i(\tilde{\theta}, x) &= 0 \\ \nabla_\theta F(\tilde{\theta}, \tilde{\lambda}, \tilde{\nu}, x) &= 0. \end{aligned} \tag{19}$$

We will reduce the KKT equation to an algebraic equation. We first let $z = (\theta, \lambda, \nu)$ for ease of notation and then define the function

$$g(z, x) = \begin{bmatrix} \nabla_\theta F(\theta, \lambda, \nu, x) \\ \mathbf{diag}(\lambda) f(\theta, x) \\ h(\theta, x)] \end{bmatrix} \tag{20}$$

where $\mathbf{diag}(\cdot)$ denotes the diagonalisation operation. We can then define the (partial) Jacobian

$$D_z g(\tilde{z}, x) = \begin{bmatrix} D_\theta \nabla_\theta F(\tilde{\theta}, \tilde{\lambda}, \tilde{\nu}, x) & D_\theta f(\tilde{\theta}, x)^T & D_\theta h(\tilde{\theta}, x)^T \\ \mathbf{diag}(\tilde{\lambda}) D_\theta f(\tilde{\theta}, x) & \mathbf{diag} f(\tilde{\theta}, x) & 0 \\ D_\theta h(\tilde{\theta}, x) & 0 & 0 \end{bmatrix} \tag{21}$$

and the (partial) Jacobian

$$D_x g(\tilde{z}, x) = \begin{bmatrix} D_x \nabla_\theta F(\tilde{\theta}, \tilde{\lambda}, \tilde{\nu}, x) \\ \mathbf{diag}(\tilde{\lambda}) D_x f(\tilde{\theta}, x) \\ D_x h(\tilde{\theta}, x) \end{bmatrix}. \tag{22}$$

The vectors $\tilde{\theta}$, $\tilde{\lambda}$, and $\tilde{\nu}$ are optimal if and only if $g(\tilde{z}, x) = 0$ with Jacobian satisfying

$$D_x s(x) = -D_z g(\tilde{\theta}, \tilde{\lambda}, \tilde{\nu}, x)^{-1} D_x g(\tilde{\theta}, \tilde{\lambda}, \tilde{\nu}, x). \tag{23}$$

References

[1] Y. Wang *et al.*, "A three-layered graph-based learning approach for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6020–6034, Oct. 2016.

[2] L. Gómez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

[3] S. Chaib, H. Liu, Y. Gu, and H. Yao, "Deep feature fusion for VHR remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4775–4784, Aug. 2017.

[4] N. Longbotham, C. Chaapel, L. Bleiler, C. Padwick, W. J. Emery, and F. Pacifici, "Very high resolution multiangle urban classification analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1155–1170, Apr. 2012.

[5] W. Zhao and S. Du, "Learning multiscale and deep representations for classifying remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 113, pp. 155–165, Mar. 2016.

[6] F. Ghazouani, I. R. Farah, and B. Solaiman, "A multi-level semantic scene interpretation strategy for change interpretation in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8775–8795, Nov. 2019.

[7] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.

[8] Q. Zhu, Y. Zhong, L. Zhang, and D. Li, "Adaptive deep sparse semantic modeling framework for high spatial resolution image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6180–6195, Oct. 2018.

[9] Q. Zhu, C. Liao, H. Hu, X. Mei, and H. Li, "MAP-Net: Multiple attending path neural network for building footprint extraction from remote sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6169–6181, Jul. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

[10] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.

[11] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1097–1105.

[12] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[13] X. Ma, H. Wang, and J. Wang, "Semisupervised classification for hyperspectral image based on multi-decision labeling and deep feature learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 120, pp. 99–107, Oct. 2016.

[14] Z. Wu, Y. Li, A. Plaza, J. Li, F. Xiao, and Z. Wei, "Parallel and distributed dimensionality reduction of hyperspectral data on cloud computing architectures," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2270–2278, Jun. 2016.

[15] L. Chen, J. Xiao, P. Zou, and H. Li, "Lie to me: A soft threshold defense method for adversarial examples of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[16] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, May 2021.

[17] J. Peng *et al.*, "Overcoming long-term catastrophic forgetting through adversarial neural pruning and synaptic consolidation," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 12, 2021, doi: 10.1109/TNNLS.2021.3056201.

[18] W. Li *et al.*, "Stacked Autoencoder-based deep learning for remote-sensing image classification: A case study of African land-cover mapping," *Int. J. Remote Sens.*, vol. 37, no. 23, pp. 5632–5646, 2016.

[19] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[20] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Feb. 2017.

[21] A. Comber, P. Fisher, C. Brunsdon, and A. Khmag, "Spatial analysis of remote sensing image classification accuracy," *Remote Sens. Environ.*, vol. 127, pp. 237–246, Dec. 2012.

[22] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.

[23] H. Li *et al.*, "RS-MetaNet: Deep metametric learning for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6983–6994, Aug. 2021.

[24] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1126–1135.

[25] Z. Li, F. Zhou, F. Chen, and H. Li, "Meta-SGD: Learning to learn quickly for few-shot learning," 2017, *arXiv:1707.09835*.

[26] J. Lu, P. Gong, J. Ye, and C. Zhang, "Learning from very few samples: A survey," 2020, *arXiv:2009.02653*.

[27] H. Peng, "A comprehensive overview and survey of recent advances in meta-learning," 2020, *arXiv:2004.11149*.

[28] X. Sun, B. Wang, Z. Wang, H. Li, H. Li, and K. Fu, "Research progress on few-shot learning for remote sensing image interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2387–2402, 2021.

[29] Y. Chen, X. Wang, Z. Liu, H. Xu, and T. Darrell, "A new meta-baseline for few-shot learning," 2020, *arXiv:2003.04390*.

[30] J. Snell, K. Swersky, and R. S. Zemel, "Prototypical networks for few-shot learning," 2017, *arXiv:1703.05175*.

[31] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.

[32] S. Ravi and H. Larochelle, "Optimization as a model for few-shot learning," in *Proc. Int. Conf. Learn. Represent.*, 2016.

[33] V. Garcia and J. Bruna, "Few-shot learning with graph neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2017.

[34] J. Vanschoren, "Meta-learning: A survey," 2018, *arXiv:1810.03548*.

[35] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," 2020, *arXiv:2004.05439*.

[36] G. Koch *et al.*, "Siamese neural networks for one-shot image recognition," in *Proc. ICML Deep Learn. Workshop*, vol. 2, 2015.

[37] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3630–3638.

[38] Z. Wu, Y. Li, L. Guo, and K. Jia, "PARN: Position-aware relation networks for few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6659–6667.

[39] B. Hariharan and R. Girshick, "Low-shot visual recognition by shrinking and hallucinating features," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3018–3027.

[40] S. Yang, L. Liu, and M. Xu, "Free lunch for few-shot learning: Distribution calibration," in *Proc. Int. Conf. Learn. Represent.*, 2021.

[41] X. Zhang *et al.*, "Spectral–spatial fractal residual convolutional neural network with data balance augmentation for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10473–10487, Dec. 2021.

[42] Z. Bao, Y.-X. Wang, and M. Hebert, "Bowtie networks: Generative modeling for joint few-shot recognition and novel-view synthesis," *IEEE Trans. Geosci. Remote Sens.*, 2020.

[43] M. Zhai, H. Liu, and F. Sun, "Lifelong learning for scene recognition in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1472–1476, Sep. 2019.

[44] G. Tseng, H. Kerner, C. Nakalembe, and I. Becker-Reshef, "Learning to predict crop type from heterogeneous sparse labels using meta-learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1111–1120.

[45] M. Ruswurm, S. Wang, M. Korner, and D. Lobell, "Meta-learning for few-shot land cover classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 200–201.

[46] A. Nichol, J. Achiam, and J. Schulman, "On first-order meta-learning algorithms," 2018, *arXiv:1803.02999*.

[47] C. Ma, X. Mu, P. Zhao, and X. Yan, "Meta-learning based on parameter transfer for few-shot classification of remote sensing scenes," *Remote Sens. Lett.*, vol. 12, no. 6, pp. 531–541, Jun. 2021.

[48] L. Zhou, P. Cui, S. Yang, W. Zhu, and Q. Tian, "Learning to learn image classifiers with visual analogy," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11497–11506.

[49] A. Antoniou, H. Edwards, and A. Storkey, "How to train your MAML," 2018, *arXiv:1810.09502*.

[50] J. X Wang *et al.*, "Learning to reinforcement learn," 2016, *arXiv:1611.05763*.

[51] Y. Duan, J. Schulman, X. Chen, P. L. Bartlett, I. Sutskever, and P. Abbeel, "Fast reinforcement learning via slow reinforcement learning," 2016, *arXiv:1611.02779*.

[52] A. Santoro, S. Bartunov, M. Botvinick, D. Wierstra, and T. Lillicrap, "Meta-learning with memory-augmented neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1842–1850.

[53] Y. Li, Z. Shao, X. Huang, B. Cai, and S. Peng, "Meta-FSEO: A meta-learning fast adaptation with self-supervised embedding optimization for few-shot remote sensing scene classification," *Remote Sens.*, vol. 13, no. 14, p. 2776, Jul. 2021.

[54] L. Li, J. Han, X. Yao, G. Cheng, and L. Guo, "DLA-MatchNet for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7844–7853, Sep. 2021.

[55] P. Zhu *et al.*, "Projection learning with local and global consistency constraints for scene classification," *ISPRS-J. Photogramm. Remote Sens.*, vol. 144, pp. 202–216, Oct. 2018.

[56] B. N. Oreshkin, P. Rodriguez, and A. Lacoste, "TADAM: Task dependent adaptive metric for improved few-shot learning," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Red Hook, NY, USA: Curran Associates, 2018, pp. 719–729.

[57] S. Barratt, "On the differentiability of the solution to convex optimization problems," 2018, *arXiv:1804.05098*.

[58] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[59] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst. (GIS)*, 2010, pp. 270–279.

[60] G. S. Xia, J. Hu, F. Hu, B. Shi, X. Bai, Y. Zhong, L. Zhang, and X. Lu, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Apr. 2017.

[61] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.

[62] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Artif. Intell. Statist.*, 2011, pp. 315–323.
[63] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.



Zhenqi Cui is currently pursuing the Ph.D. degree with Central South University, Changsha, China.

His research interests include remote sensing image understanding and machine learning.

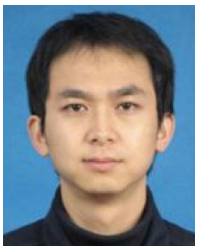


Wang Yang received the bachelor's degree in artificial intelligent science and technology from Central South University, Changsha, China in 2021.



Li Chen received the B.Sc. and M.Sc. degrees from the School of Software, Central South University, Changsha, China, in 2015 and 2018, respectively. He is currently pursuing the Ph.D. degree with the School of Geosciences and Info-Physics, Central South University.

His research interests include transfer learning, deep learning, and visualization.



Haifeng Li (Member, IEEE) received the master's degree in transportation engineering from the South China University of Technology, Guangzhou, China, in 2005, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2009.

He was a Research Associate with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong, in 2011, and a Visiting Scholar with the University of Illinois at Urbana–Champaign, Urbana, IL, USA, from 2013 to 2014. He is currently a Professor with the School of Geosciences and Info-Physics, Central South University, Changsha, China. He has authored over 30 journal articles. His research interests include geo/remote sensing big data, machine/deep learning, and artificial/brain-inspired intelligence.

Dr. Li is a reviewer for many journals.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:45:42 UTC from IEEE Xplore. Restrictions apply.